

Tejal Kulkarni

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EDUCATION

UNIVERSITY OF CALIFORNIA SAN DIEGO | *M.S. in Computer Science* | GPA: 4.0/4.0 2025 – 2027

- Coursework: Recommendation Systems, Introduction to Robotics, Generative AI, AI Agents, Data Systems for ML, Advanced Computer Vision

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *B.Tech in Computer Science* | GPA: 9.44/10 2021 – 2025

- Coursework: Foundations of ML, Deep Learning, Computer Vision, Reinforcement Learning, Convex Optimization, Information Theory, Theory of Computation, Compilers, Database Management Systems, Operating Systems, Computer Networks, Cryptology, Topics in NLP, Data Mining, Tensor Techniques & Algorithms

PUBLICATIONS

RoboTALES: Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures 2026

H. Gani, T. Kulkarni, M. V. Chodavarapu, N. Hansen, M. Chandraker. [ECCV 2026]

Foundation Model Priors Enhance Object Focus in Feature Space for Source-Free Object Detection 2026

S. VCR, R. Lalla, A. Dayal, T. Kulkarni, A. Lalla, V. N. Balasubramanian, M. H. Khan. [CVPR 2026]

On Evaluation of Vision Datasets and Models using Human Competency Frameworks 2024

R. Ramachandran, T. Kulkarni, C. Sharma, D. Vijaykeerthy, V. N. Balasubramanian. [ICML 2024 DMLR Workshop]

RESEARCH EXPERIENCE

UNIVERSITY OF CALIFORNIA, SAN DIEGO | *Graduate Researcher* San Diego, CA

RoboTALES: Learning Reasoning-Guided Robot Policies via Task-Aligned Simulated Futures 2025 – 2026

- Developed a **diffusion-based** world model guided by LLM and VLM reasoning to improve long-horizon planning and action generation.
- Implemented a **planner-critic framework** with RL to align predicted future states with task instructions; demonstrated improved rollout consistency on **RoboCasa** and **LIBERO** benchmarks.

UMASS HUMAN-CENTERED ROBOTICS LAB | *Graduate Researcher*

Remote

Zero-Shot Generalizable Multi-Task World Models

Feb. 2026 – present

- Built single world model jointly trained on **71 visual control tasks** (OGBench, Meta-World, RoboDesk; over a frozen **DINOv2** backbone with a universal action encoder.
- Added dual natural-language task/robot conditioning via frozen **MiniLM** sentence embeddings, enabling zero-shot transfer to new tasks/embodyments without retraining; matched LeWM single-task specialist on cube within 3 SR points at **~18× less compute**.

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *Undergraduate Researcher*

Hyderabad, India

Foundation Model Priors Enhance Object Focus in Feature Space for Source-Free Object Detection

2024 – 2025

- Designed a **novel loss function** to mitigate limitations of current SOTA source-free domain adaptation (SFDA) object detection methods.
- Leveraged **Grounded-SAM** (foundation model) to reduce feature distortions from domain shifts; validated across multiple SFDA benchmarks.

INDIAN INSTITUTE OF TECHNOLOGY HYDERABAD | *Undergraduate Researcher*

Hyderabad, India

On Evaluation of Vision Datasets and Models using Human Competency Frameworks

2023 – 2024

- Assessed model calibration and compared vision dataset complexities leveraging **Item Response Theory (IRT)** – a human competency framework.
- Extracted IRT latent parameters across 91 vision models on ImageNet; showed a subset of just **10 discriminable images** ranks models with **Kendall correlation of 0.85** vs. full-set ranking.

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN | *REU Fellow, NCSA*

Champaign, IL

Deep Sea Video Classification

2023

- Built a **CNN-LSTM-based model** to detect the breathing state of sea turtles from deep-sea video.
- Developed modular ML features for the **open-source DRYML** library, focusing on reproducibility, scalability, and deployment-readiness.

WORK EXPERIENCE

IBM | *Machine Learning Engineering Intern*
WatsonX Assistant Text-to-Speech Team

San Jose, CA
2026

- Built an **in-house emotion classifier** to auto-label existing speech corpora with emotion annotations and extended their production model to include emotion steering capabilities.

QUALCOMM INSTITUTE, UC SAN DIEGO | *Graduate Researcher*
A Human-in-the-Loop AI Memoir System

San Diego, CA
2025 – present

- Built a **persona-grounded conversational AI** simulating a real person's voice and memory using **RAG** over personal archives, with a **multi-agent Judge–Corrector** pipeline on to ensure factual accuracy
- Developed an **AI PhD advisor** discovery and simulation platform that scores student–advisor research fit using semantic similarity and **information-theoretic models**, and recreates committee reviews via a **multi-agent discussion engine** backed by live academic paper ingestion

QUALCOMM | *Software Engineering Intern*
Build Pipeline Optimization

Hyderabad, India
2024 | 10 Weeks

- Optimized large-scale build pipelines by implementing and configuring **ccache**, significantly reducing compilation duration and developer turnaround time.

ACADEMIC PROJECTS

Multi Agent Robot Navigation *Advanced Computer Vision*

2026

- Built a training-free **5-agent pipeline** (Perception, Mapping, Planning, Critic, Action) for object-goal navigation in **ProcTHOR** and achieved **77.78% SR** vs. 41.67% single-VLM baseline.

Efficient Few-Shot Style Adaptation for T2I Generation Models *Computer Vision*

2025

- Implemented **LoRA** and **hybrid Textual Inversion + LoRA** fine-tuning on Stable Diffusion for few-shot T2I generation, reducing overfitting and improving stylistic consistency.
- Explored DreamBooth and StyleDrop with **CLIP-based evaluation** (FID, CLIP similarity); built scalable pipelines using **Hugging Face Diffusers** and **OpenCLIP**.

Legal Document Simplification *Advanced Topics in NLP*

2024

- Developed legal text simplification models with **BART/LongT5**, boosting SARI from **35.3** → **45.4** by addressing token-limit constraints.
- Built an **extract-then-simplify pipeline** (SummaRuNNer + LLM fine-tuning) on the MILDSum dataset; explored chunking, summarization, and rhetorical role embeddings for long-document understanding.

HONORS & AWARDS

- **1st Place**, Idea Quest – Edge AI Hackathon organized for Qualcomm Interns 2024
- **All India Rank 990** in IIT-JEE Advanced (155K candidates) 2021
- Among the **Top 300** students selected for **InMO**, qualifying round for the **IMO** 2018
- Scholarship recipient, **Mathematics Prodigy Competition** 2017

TEACHING & ACTIVITIES

- **Teaching Assistant**, Foundations of Machine Learning, Theory of Computation – IIT Hyderabad 2024, 2025
- **Organized and led** a Computer Vision workshop at Tinkerer's Lab, mentoring students on CV concepts 2024
- **Machine Learning Core**, AI and Robotics Club (Tinkerer's Lab), IIT Hyderabad 2024
- Represented IIT Hyderabad at the **Inter-IIT Sports Meet** hosted by IIT Delhi 2022

TECHNICAL SKILLS

Languages Python, C/C++, Java, JavaScript, Bash, Sed, Awk

ML & Systems PyTorch, TensorFlow, OpenCV, CUDA, Docker, ROS2, OpenGL, MediaPipe, Pandas, scikit-learn, SciPy, CLIP, Diffusers, MuJoCo

Libraries NumPy, LangChain, Langgraph, unsloth, SpaCy, NLTK, pgvector, Streamlit, ReactJS, NodeJS, Express, FastAPI, Material UI, socket programming, AWS (Bedrock, boto3)